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LAB 4: Consuming APIs in flutter

Q. Which dependencies did you use and why?

= is used .

* http

the reason why I used http is to communicate with the API (send and receive data)

flutter doesn't directly fetch data from the internet without a package. the http package allows us to fetch posts (GET); create posts (POST), update posts (PUT/PATCH) and delete posts (DELETE)

ex: - fetch post from api

- ~~create~~ send new post data to server.

* provider / setState (state management)

this helps to manage application state because when data changes (like after adding or deleting a post), the UI must update automatically

setState() .: for small apps

provider : . for better state management.

* dart: Convert:

it helps to convert json data into Dart objects
where API returns data in json format, so we
need to decode it.

ex: Convert API Response to Dart Map/List.

(2) How can you handle API Exceptions?

I used:

a) Try-catch-Block

it is used to catch errors during API calls.

b) Checking response status code

after calling APIs, check if request was successful.
where common codes are:

200 → Success

201 → created successfully

404 → Not found

500 → server error.

c) Handling network errors

it happens when no internet and slow connection
this is solved by showing message "no internet connection"

d) Handling invalid data

if API returns unexpected data;

• use validation before displaying.

e) User feedback:

it always inform user Loading indicator (CircularProgressIndicator), error messages (SnackBar / Text).

(3) Explain the Future Widget and the UI part that depends on it

- What is Future is;

A future represents a value that will be available later

Used when fetching data from API (asynchronous operation)

Future Widget Used: Future Builder

The reason why we need Future Builder is because it helps build UI based on API data and it automatically update UI when data is Loading, Loaded and error.

How it works

it works in 3 main states:

(1) Loading state

while waiting for data where UI shows

CircularProgressIndicator

(2) Success state

when data is received and UI shows

List of posts (ListView)

(3) Error state

if something goes wrong and UI shows

Error message

* part of UI depends on future?

the post List Screen depends on future because it waits for API data before displaying posts.

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assignment-4 > lib > main.dart > ...

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2
3 import 'package:connectivity_plus/connectivity_plus.dart';
4 import 'package:flutter/foundation.dart';
5 import 'package:flutter/material.dart';
6 import 'package:hive_flutter/hive_flutter.dart';
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8
9 Run | Debug | Profile
10 void main() async {
11   WidgetsFlutterBinding.ensureInitialized();
12   await Hive.initFlutter();
13   await Hive.openBox('cache');
14   runApp(const MyApp());
15 }
16
17 class MyApp extends StatelessWidget {
18   const MyApp({super.key});
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\unipod\Desktop\dart-school-lab\assignment-4> flutter pub get

PS C:\Users\unipod\Desktop\dart-school-lab\assignment-4> flutter run -d windows

Launching lib\main.dart on Windows in debug mode...

Building Windows application... 11.3s

✓ Built build\windows\x64\runner\Debug\flutter_application_1.exe

Syncing files to device Windows... 189ms

Flutter run key commands.

- r Hot reload.
- R Hot restart.
- h List all available interactive commands.
- d Detach (terminate "flutter run" but leave application running).
- c Clear the screen
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flutter_application_1

Posts Manager

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